

ViCO: Vision-Augmented Time Series Anomaly Detection with Prototype-Guided Coordination

Xuesong Zeng
xuesong.zeng@csu.edu.cn
Central South University

Mingjun Cao
cmjcat@csu.edu.cn
Central South University

Hao Miao*
hao.miao@polyu.edu.hk
The Hong Kong Polytechnic
University

Chenxi Liu
chenxi.liu@cair-cas.org.hk
Hong Kong Institute of Science &
Innovation, Chinese Academy of
Sciences

Senzhang Wang*
szwang@csu.edu.cn
Central South University

Abstract

Time series anomaly detection aims to identify time points or intervals that deviate from normal temporal behavior. In reconstruction-based detection, this requires errors that distinguish anomalies rather than merely accurate reconstruction. However, most existing methods rely solely on temporal modeling, while vision-augmented approaches focus primarily on forecasting, leaving the potential of structural visual cues for reconstruction-based detection underexplored. Moreover, bridging temporal and visual representations further remains challenging. To address these challenges, we propose a Vision-Augmented Coordination framework for time series anomaly detection, called ViCO. A decoupled visual teacher and temporal student characterize normal behavior from complementary views: tri-view visual representations capture structural regularities, whereas raw sequences model value-sensitive dynamics. Prototype-Guided Coordination retrieves normal structural prototypes and conditions the temporal decoder, translating visual structure into reconstruction guidance for anomaly scoring. Extensive experiments on five real-world datasets demonstrate that ViCO consistently outperforms strong baselines by up to 9.0%, while enabling teacher-free inference for lightweight deployment. Code is available at <https://github.com/Cedarr1/ViCO>.

CCS Concepts

• **Information systems** → **Data mining**; • **Computing methodologies** → **Anomaly detection**.

Keywords

Time Series Anomaly Detection, Vision-Augmentation, Prototype-Guided Coordination

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1 Introduction

Time series data are widely collected in real-world systems such as fault diagnosis and fraud detection [43, 50], where time series anomaly detection (TSAD) aims to identify time points or intervals that deviate from normal temporal regularities [20, 26, 37].

Existing TSAD methods approach this problem from different perspectives and can be broadly categorized into distance-based [6], density-based [19], forecasting-based [49] and reconstruction-based approaches [34]. Given scarce anomaly annotations, reconstruction-based TSAD has become a mainstream paradigm, where models are trained on normal sequences and anomalies are indicated by large reconstruction residuals between the input and its reconstruction. In this paradigm, effective anomaly scoring requires residuals to reflect departures from normality. Nevertheless, existing methods largely rely on single-modality temporal modeling, limiting their ability to characterize heterogeneous temporal structures, including local transitions, contextual organization, and sequence-level variations [4, 20]. Although vision augmentation provides complementary structural views [10, 51], current time-series-to-image designs remain mostly forecasting-oriented and primarily preserve trajectory information for future-value prediction [36, 45]. For reconstruction-based TSAD, visual information should instead complement value-sensitive temporal modeling with structure-aware knowledge aligned with anomaly-oriented reconstruction, rather than serving as generic prediction-enhancing features. This motivates a vision-augmented TSAD framework that models normal behavior from complementary visual-structural and temporal-value perspectives, and coordinates visual knowledge with temporal reconstruction so that residuals can better reflect normality violations.

However, building such a vision-augmented TSAD framework is non-trivial due to two challenges. First, it is difficult to model normal behavior from both structural and value-level perspectives

*Hao Miao and Senzhang Wang are corresponding authors.



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without conflating their roles in reconstruction. Temporal behavior exhibits heterogeneous structures, including short-term dynamics, temporal context, and recurring sequential patterns, requiring both structure-aware contextual understanding and value-sensitive temporal perception [21]. Thus temporal models based only on raw values may struggle to characterize such structures, while directly coupling structural pattern learning with value-level reconstruction may blur their distinct roles. Therefore, effective normal-pattern modeling should preserve complementary visual-structural and temporal-value perspectives, while keeping temporal reconstruction as the basis for anomaly scoring. Second, even when visual structural cues are available, adapting them to temporal reconstruction remains challenging due to the temporal–visual modality gap [7]. Visual cues are typically structure-oriented and window-level, whereas temporal features preserve ordered value dynamics for time-step-level reconstruction and anomaly-score computation. Effective adaptation, therefore, needs to determine which visual cues are relevant to the current temporal context and how strongly they should calibrate value-level reconstruction. Without such selective and constrained adaptation, visual cues may provide irrelevant or misaligned guidance, causing reconstruction errors to reflect cross-modal mismatch rather than departures from normality.

To address these challenges, we propose **ViCO**, a vision-augmented coordination framework for time series anomaly detection. ViCO performs vision augmentation in two phases, i.e., complementary pattern modeling and cross-modal guidance adaptation, which respectively address the effective modeling of normal patterns and their adaptation to temporal reconstruction. First, to model normal behavior from complementary perspectives, ViCO decouples structure-aware visual modeling from value-sensitive temporal reconstruction through Complementary Pattern Modeling. The visual structural branch reorganizes time-series windows into tri-view representations and uses a frozen visual teacher to capture complementary structural cues, including local deviations, contextual changes, and frequency-domain patterns, while the temporal value branch preserves value-sensitive temporal dynamics from raw sequences through period-aware multiscale encoding. Moreover, unlike existing dual-branch vision-augmentation designs [23, 32, 36, 51] that rely on visual processing during inference, this decoupled scheme enables efficient detection, where only the temporal detector and fixed prototype bank are retained at test time. Second, to adapt these references across the temporal–visual modality gap, ViCO develops Cross-modal Guidance Adaptation to transform visual structural prototypes into reconstruction guidance. Rather than directly fusing visual representations with temporal embeddings [18, 32, 51], ViCO performs context-conditioned prototype routing through query generation and guidance retrieval, enabling the temporal student to retrieve structurally relevant guidance for the current window. To coordinate the selected references with temporal reconstruction, ViCO further proposes Prototype-Guided Coordination, consisting of MaskAttention and Bounded FiLM. MaskAttention converts the structural guidance into a fine-grained support mask over decoder features, localizing where guidance should interact with temporal reconstruction. Bounded FiLM then provides constrained feature calibration, controlling how strongly the structural guidance can modulate value-level decoder features. As a result, visual structural knowledge guides temporal reconstruction in a

selective and controlled manner, yielding anomaly-sensitive reconstruction errors rather than cross-modal mismatch.

The main contributions are summarized as follows.

- We propose ViCO, a vision-augmented TSAD framework that coordinates visual structural cues with temporal reconstruction, enabling generalizable detection of heterogeneous anomalies.
- We introduce Complementary Pattern Modeling with a visual teacher–temporal student scheme, separating structure-aware visual modeling from value-sensitive temporal reconstruction to characterize normal patterns from complementary perspectives.
- We design Cross-modal Guidance Adaptation with Prototype-Guided Coordination, which formulates temporal–visual interaction as selective and constrained reference-to-reconstruction coordination across the modality gap.
- Extensive experiments on five real-world datasets demonstrate that ViCO consistently outperforms strong baselines. Further analyses verify that ViCO improves anomaly-sensitive reconstruction errors and enables efficient inference.

2 Related Work

2.1 Time Series Anomaly Detection

Time series anomaly detection aims to identify unusual patterns or outliers in time series, which plays a crucial role in various real-world applications [24, 42, 48]. Early studies include distance-based methods [5, 6] using distance measures over raw time-series points or subsequences for anomaly detection. Next, density-based methods [9, 19] estimate the density of points or subsequences in a representation space and identify samples in sparse regions as anomalies. To capture temporal dependencies, forecasting-based methods [13, 49] use historical observations to predict future values or subsequences, where forecasting errors serve as anomaly evidence. Reconstruction-based methods [38, 40] reconstruct input subsequences and use reconstruction errors as anomaly scores. Nevertheless, existing TSAD methods mainly rely on single-modality temporal modeling, limiting their ability to characterize heterogeneous temporal patterns in real-world scenarios [4, 20].

2.2 Vision-augmented Time Series Analytics

Visual representations can complement 1D temporal modeling by exposing structural patterns [36] and enabling the transfer of pre-trained vision knowledge [25] to time-series analysis. Existing studies can be broadly grouped into two paradigms. Direct visual modeling methods [45] transform time series into images and directly process them with pre-trained vision backbones or vision-language models. Cross-modal alignment methods [41, 51] bridge temporal and visual representations through cross-modal interaction. Among them, retrieval-based interaction [11, 18] and knowledge distillation [17, 23] are particularly relevant to our work, as they enable temporal models to incorporate useful visual guidance without heavy online inference cost. In this work, we focus on the cross-modal alignment paradigm. However, existing time-series-to-image designs are mostly forecasting-oriented and primarily preserve trajectory information for future-value prediction [10, 51], whereas reconstruction-based TSAD requires visual information aligned with anomaly-oriented reconstruction.

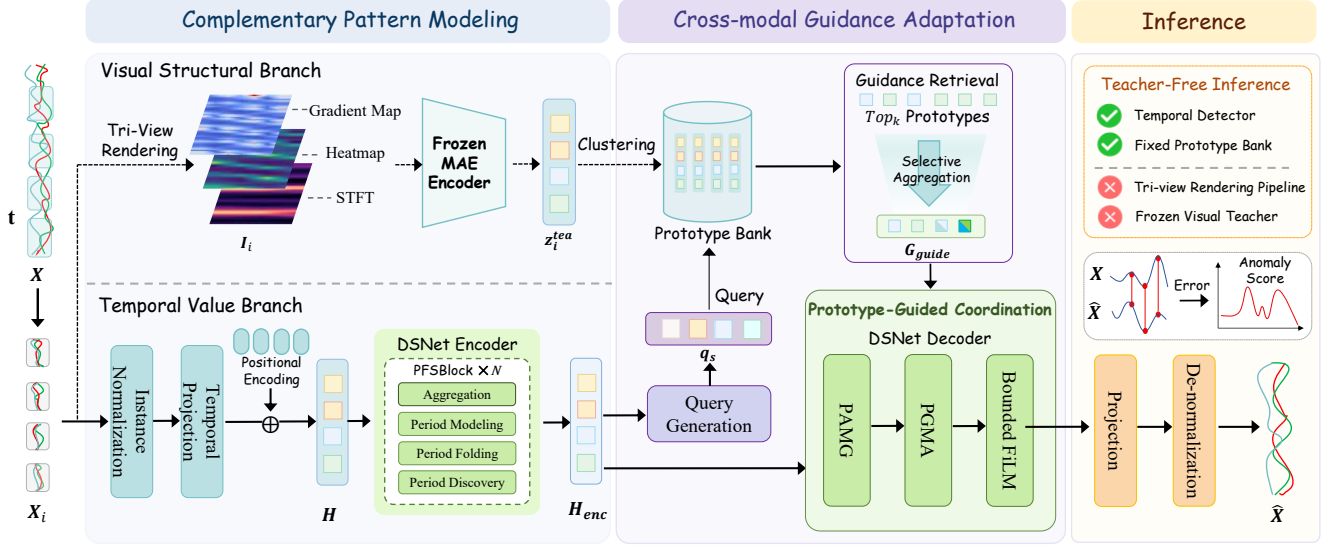


Figure 1: ViCO framework overview.

3 Problem Definition

Definition 3.1 (Time Series). A time series is denoted by $\mathcal{X} = \langle \mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T \rangle$, where $\mathbf{x}_t \in \mathbb{R}^C$ is the observation at time step t . If $C = 1$, $\mathcal{X}$ is univariate; otherwise, it is multivariate.

Vision-Augmented Time Series Anomaly Detection. Given a normal training set $\mathcal{D}_{tr}$, we segment time series into sliding-window samples $\mathcal{W}_{tr} = \{\mathbf{X}_i\}_{i=1}^N$, where $\mathbf{X}_i \in \mathbb{R}^{L \times C}$, L is the window length, and N is the number of windows. The goal is to learn a reconstruction-based anomaly detector $F_\theta(\cdot; \mathcal{P})$, where $\mathcal{P}$ is a prototype bank constructed from visual representations of normal training samples. The visual teacher provides training-time structural knowledge only, while anomaly detection is performed by the temporal detector with the fixed prototype bank.

The optimal detector is obtained by solving

$$\theta^* = \arg \min_{\theta} \mathcal{L}_{train}(\theta; \mathcal{W}_{tr}, \mathcal{P}), \quad (1)$$

where $\mathcal{L}_{train}$ denotes the training objective of ViCO.

For a reconstructed window $\hat{\mathbf{X}}_i = F_{\theta^*}(\mathbf{X}_i; \mathcal{P})$, the anomaly score at local position $u = 1, \dots, L$ is defined by the reconstruction error:

$$s_{i,u} = s_{i,u}^{rec} = \mathcal{A}_{ch} \left(\left\{ \left(X_{i,u,c} - \hat{X}_{i,u,c} \right)^2 \right\}_{c=1}^C \right), \quad (2)$$

where $\mathcal{A}_{ch}(\cdot)$ aggregates the reconstruction errors across variables. The time point at local position u is identified as anomalous when $s_{i,u} > \gamma$, where γ is a detection threshold.

4 Methodology

Following Section 3, we use sliding windows as the basic modeling units. Given a time series $\mathcal{X} = \langle \mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T \rangle$, the i -th window is denoted as $\mathbf{X}_i = [\mathbf{x}_{\tau_i}, \mathbf{x}_{\tau_i+1}, \dots, \mathbf{x}_{\tau_i+L-1}] \in \mathbb{R}^{L \times C}$, where $\tau_i = 1 + (i-1)S$, L is the window length, and S is the stride. ViCO takes $\mathbf{X}_i$ as input and reconstructs it as $\hat{\mathbf{X}}_i = F_\theta(\mathbf{X}_i; \mathcal{P})$, where $\mathcal{P}$ is the fixed normal structural prototype bank. The reconstruction discrepancy between $\mathbf{X}_i$ and $\hat{\mathbf{X}}_i$ provides the anomaly score in Eq. (2). The

following subsections introduce the framework overview, the two modeling phases, and the overall objective.

4.1 Framework Overview

Figure 1 shows the overall framework of ViCO. Given a window $\mathbf{X}_i$, ViCO follows two phases to coordinate visual structural knowledge with temporal reconstruction.

In the first phase, complementary pattern modeling decouples normal-pattern learning into a visual structural branch and a temporal value branch. As shown in Figure 1, the visual branch maps the rendered image I_i to the teacher embedding z_i^{tea} and builds a fixed prototype bank $\mathcal{P} = \{\mathbf{p}_k\}_{k=1}^K$, while the temporal branch encodes raw values into $\mathbf{H}_{enc,i}$ for reconstruction.

In the second phase, cross-modal guidance adaptation retrieves structural references and adapts them as guidance for temporal reconstruction. As shown in Figure 1, $\mathbf{H}_{enc,i}$ is summarized into a query $\mathbf{q}_{s,i}$ to retrieve guidance $\mathbf{G}_{guide,i}$ from $\mathcal{P}$, which is then coordinated with the DSNet decoder to produce $\hat{\mathbf{X}}_i$. Concretely, Prototype-Guided Coordination is inserted after DSNet’s anomaly-attention block and before its final linear reconstruction head, where it adapts the decoder state with retrieved structural guidance before projection back to the variable space.

During inference, the rendering pipeline and the visual teacher are removed. ViCO retains only the temporal detector and the fixed prototype bank $\mathcal{P}$, enabling efficient anomaly detection.

4.2 Complementary Pattern Modeling

The first phase of ViCO characterizes normal temporal patterns from complementary visual and temporal perspectives under a role-separated design. Rather than treating the visual teacher and temporal student as parallel reconstruction branches, ViCO assigns structure-aware reference construction to the visual branch and value-level reconstruction to the temporal branch. This phase includes tri-view rendering, normal prototype construction, and temporal student encoding.

4.2.1 Visual structural branch. The visual structural branch constructs structural knowledge from rendered time-series images. Given a normal window X_i , it first produces a tri-view representation I_i , then feeds it into a frozen MAE teacher to obtain the teacher embedding z_i^{tea} . The teacher embeddings are finally clustered into the fixed prototype bank $\mathcal{P}$.

Tri-View Rendering. Unlike forecasting-oriented renderings that primarily preserve raw trajectory shapes, reconstruction-based TSAD requires representations that characterize normal temporal behavior and distinguish it from deviations for anomaly scoring [10]. Following the taxonomy of point, contextual, and collective anomalies [4], we therefore design Tri-view representations that encode local transitions, period-organized value context, and spectral regularities through the Gradient, Heatmap, and STFT views, respectively. Together, these views provide complementary normal cues for anomaly-sensitive reconstruction.

Let $X_i \in \mathbb{R}^{L \times C}$ denote an input window, where L is the window length and C is the number of variables. Unless otherwise specified, we use rendering period $P_r = 24$ and normalize each variable along the temporal axis as $\tilde{X}_i^{(c)} = \eta(X_i^{(c)} - \mu_i^{(c)})/(\sigma_i^{(c)} + \epsilon)$, where $\eta = 0.4$, and $\mu_i^{(c)}$ and $\sigma_i^{(c)}$ denote the temporal mean and standard deviation of the c -th variable, respectively. Let $\mathcal{F}_{P_r}(\cdot)$ denote the period-aware folding operator, $\mathcal{B}(\cdot)$ bilinear resizing to 224×224 , and $\mathcal{N}_{01}(\cdot)$ per-view min-max normalization. Variable-wise maps are averaged to form one channel per view before the three channels are stacked in a fixed order. The tri-view representation is formed by stacking the three views along the channel dimension:

$$I_i = \text{Concat}(\mathbf{V}_{\text{spec}}, \mathbf{V}_{\text{heat}}, \mathbf{V}_{\text{grad}}), \quad (3)$$

where $\mathbf{V}_{\text{spec}}$, $\mathbf{V}_{\text{heat}}$, and $\mathbf{V}_{\text{grad}}$ denote the STFT view, the heatmap view, and the gradient view, respectively. For presentation clarity, we introduce the three views in the order of Gradient, Heatmap, and STFT, while the actual channel concatenation follows the implementation order of STFT, Heatmap, and Gradient.

Gradient View. The gradient view complements value-level rendering by highlighting abrupt local changes. For each variable, we compute first-order temporal differences, set the first gradient value to zero, and apply zero padding to complete the period-aware folded layout. The resulting gradient magnitudes are then rendered as

$$\mathbf{V}_{\text{grad}} = \mathcal{N}_{01} \left(\frac{1}{C} \sum_{c=1}^C \left| \mathcal{B}(\mathcal{F}_{P_r}(\nabla \tilde{X}_i^{(c)})) \right| \right). \quad (4)$$

Here, $\nabla \tilde{X}_i^{(c)}$ denotes the first-order temporal difference with zero initialization at the first time step. This view makes short-range transitions salient, providing normal references that are particularly informative for point-wise deviations.

Heatmap View. The heatmap view preserves contextual value organization under a period-aware two-dimensional layout. We first pad the sequence to length $L' = \lceil L/P_r \rceil P_r$ and then apply $\mathcal{F}_{P_r}$ to reshape the sequence into a matrix of size $P_r \times (L'/P_r)$. Averaging the resized maps across variables gives

$$\mathbf{V}_{\text{heat}} = \mathcal{N}_{01} \left(\frac{1}{C} \sum_{c=1}^C \mathcal{B}(\mathcal{F}_{P_r}(\tilde{X}_i^{(c)})) \right). \quad (5)$$

By aligning values at comparable within-period positions, this view makes contextual departures more visible.

STFT View. The STFT view is introduced to represent frequency-domain regularities that are difficult to capture directly from the time-domain curve. We use a Hann window with requested window size 64, hop length 16, and FFT size 128; the implementation caps these values by the input length, yielding an effective FFT size of 64 for $L = 96$. For each variable, we compute a log-magnitude short-time Fourier transform map and then average the resulting spectrograms across variables:

$$\mathbf{V}_{\text{spec}} = \mathcal{N}_{01} \left(\mathcal{B} \left(\frac{1}{C} \sum_{c=1}^C \log(1 + |\text{STFT}(\tilde{X}_i^{(c)})|) \right) \right). \quad (6)$$

This view captures spectral regularities and complements the time-domain views with sequence-level frequency information.

Together, the three views provide a informative visual summary of each time-series window, allowing the visual teacher to encode complementary temporal regularities from texture-like visual patterns. Such renderings are also compatible with MAE-style patch encoding, where the frozen visual teacher provides stable structural representations without relying on semantic image categories [23].

Normal Prototype Construction. After tri-view rendering, we use offline K-means clustering to group the visual embeddings of normal training windows into a fixed prototype bank. Each rendered image I_i is first normalized with ImageNet statistics and encoded by a frozen MAE teacher $f_T(\cdot)$ to obtain a consistent visual representation. Specifically, let $U_i^{\text{tea}} = f_T(I_i) \in \mathbb{R}^{N_p \times d_T}$ denote the output token sequence from the last transformer block, where N_p is the number of visual patches and d_T is the teacher feature dimension. The window-level teacher embedding is obtained by mean pooling:

$$z_i^{\text{tea}} = \frac{1}{N_p} \sum_{n=1}^{N_p} U_{i,n}^{\text{tea}}. \quad (7)$$

Freezing the teacher keeps the visual embedding space fixed, ensuring that the constructed prototypes remain consistent structural references during student training.

Given the teacher embeddings of normal training windows, $\mathcal{Z}_v = \{z_i^{\text{tea}}\}_{i=1}^N$, we apply K-means clustering to construct K normal structural prototypes:

$$\{\mathbf{p}_k\}_{k=1}^K = \arg \min_{\{\mathbf{c}_k\}_{k=1}^K} \sum_{i=1}^N \min_{1 \leq k \leq K} \|z_i^{\text{tea}} - \mathbf{c}_k\|_2^2. \quad (8)$$

We define the resulting fixed prototype bank as

$$\mathcal{P} = \{\mathbf{p}_1, \dots, \mathbf{p}_K\}. \quad (9)$$

Each prototype represents a centroid in the teacher embedding space and summarizes a representative normal structural pattern. The resulting prototype bank $\mathcal{P}$ is kept fixed during student training and inference, serving as a compact set of normal structural references for cross-modal guidance adaptation.

4.2.2 Temporal value branch. The temporal value branch models value-sensitive dynamics from raw time-series windows and serves as the reconstruction pathway for anomaly scoring. To make reconstruction sensitive to departures from normal temporal behavior, we adopt a lightweight period-aware encoder with compact

periodic inductive biases instead of heavy sequence-wide interaction. It first embeds normalized temporal values and then applies stacked Period-Folded Separable Blocks (PFSBlocks) to produce $\mathbf{H}^{\text{enc}}$. We omit the sample index for brevity unless a batch index is explicitly required.

Temporal Value Embedding. Given $\mathbf{X} \in \mathbb{R}^{L \times C}$, the temporal branch first applies instance normalization along the temporal dimension to reduce distributional scale variations, obtaining $\mathbf{X}_{\text{norm}}$ with $\mathbf{X}_{\text{norm}}^{(c)} = (\mathbf{X}^{(c)} - \mu^{(c)})/(\sigma^{(c)} + \epsilon)$. It then maps the normalized sequence into a d -dimensional temporal embedding space and injects temporal order through positional encodings:

$$\mathbf{H}^{\text{emb}} = \text{Dropout} \left(\text{Proj}_{\text{temp}}(\mathbf{X}_{\text{norm}}) + \mathbf{E}_{\text{pos}} \right). \quad (10)$$

Here, $\mathbf{H}^{\text{emb}} \in \mathbb{R}^{L \times d}$ denotes the initial temporal representation, $\text{Proj}_{\text{temp}}(\cdot)$ denotes the convolutional temporal projection layer, and $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{L \times d}$ denotes sinusoidal positional encodings.

Period-Aware Encoding. Normal temporal behavior often exhibits recurring patterns at multiple scales, whereas modeling long-range dependencies with full self-attention incurs high computational cost. We therefore employ stacked PFSBlocks in the temporal branch to capture multi-scale periodic patterns. Each PFSBlock comprises Period Discovery, Period Folding, Period Modeling, and Aggregation, producing a period-aware representation.

Period Discovery. Let $\mathbf{H} \in \mathbb{R}^{B \times L \times d}$ denote the input to the current PFSBlock, and let $\mathbf{H}_b \in \mathbb{R}^{L \times d}$ denote the representation of the b -th sample. We first compute the batch-averaged amplitude spectrum along the temporal dimension:

$$A(f) = \frac{1}{Bd} \sum_{b=1}^B \sum_{r=1}^d |\text{FFT}(\mathbf{H}_b)_{f,r}|. \quad (11)$$

We then select the top- k_{freq} non-zero frequencies with the largest amplitudes and convert each selected frequency f_j into its corresponding period $p_j = \lfloor L/f_j \rfloor$. This step identifies dominant periodic scales for subsequent multiscale temporal modeling.

Period Folding. For each candidate period p_j , we pad the sequence to length $L'_j = \lceil L/p_j \rceil p_j$ and fold it into a period-aligned tensor through $\mathcal{F}_{p_j}(\cdot)$. This transformation rearranges repeated temporal patterns into a locally organized two-dimensional layout, making intra-period and inter-period dependencies easier to model.

Period Modeling. Each folded representation is then processed by a lightweight convolutional block and unfolded back to the original temporal resolution:

$$\tilde{\mathbf{H}}_b^{(j)} = \mathcal{F}_{p_j}^{-1} \left(\Phi_{\text{conv}}(\mathcal{F}_{p_j}(\text{Pad}_{p_j}(\mathbf{H}_b))) \right), \quad (12)$$

where $\text{Pad}_{p_j}(\cdot)$ pads the sequence to a length divisible by p_j , and $\Phi_{\text{conv}}(\cdot)$ denotes the lightweight convolutional block for period-specific modeling. This stage captures local dependencies within each period as well as interactions across repeated periods.

Aggregation. Although the candidate periods are selected from the batch-averaged spectrum, their relative importance still varies across samples. We therefore compute sample-specific aggregation weights from the FFT amplitudes of each sample. Specifically, for sample b , the channel-averaged amplitude is computed as $\bar{A}_b(f) = \frac{1}{d} \sum_{r=1}^d |\text{FFT}(\mathbf{H}_b)_{f,r}|$. The corresponding periodic components are

then fused by softmax-weighted aggregation:

$$\mathbf{H}_b^{\text{enc}} = \mathbf{H}_b + \sum_{j=1}^{k_{\text{freq}}} \frac{\exp(\bar{A}_b(f_j))}{\sum_{m=1}^{k_{\text{freq}}} \exp(\bar{A}_b(f_m))} \tilde{\mathbf{H}}_b^{(j)}. \quad (13)$$

This aggregation preserves the relative contribution of different periodic components and yields a multiscale temporal representation adapted to each input window. Stacking N_{tb} such PFSBlocks produces the final encoder representation $\mathbf{H}^{\text{enc}} \in \mathbb{R}^{L \times d}$.

4.3 Cross-modal Guidance Adaptation

Rather than directly fusing visual and temporal features, the second phase adapts structural guidance for reconstruction through query, retrieval and coordination. The resulting decoder state is projected back and de-normalized for anomaly scoring.

4.3.1 Query Generation. To obtain structural guidance relevant to the current temporal context, we summarize the temporal encoder output into a query for the fixed prototype bank. For brevity, we omit the sample index in this subsection.

Given the encoder output $\mathbf{H}^{\text{enc}} \in \mathbb{R}^{L \times d}$, we combine temporal mean pooling and max pooling to capture the global temporal tendency and salient local responses, respectively. The pooled summary is then mapped through a projection head to obtain

$$\mathbf{q}_s = \psi(\text{Mean}(\mathbf{H}^{\text{enc}}) \oplus \text{Max}(\mathbf{H}^{\text{enc}})). \quad (14)$$

The resulting query $\mathbf{q}_s$ provides a context-aware description of the current window for prototype matching.

4.3.2 Guidance Retrieval. Given $\mathbf{q}_s$, ViCO retrieves guidance from the fixed prototype bank $\mathcal{P} = \{\mathbf{p}_1, \dots, \mathbf{p}_K\}$ through full-bank-normalized top- k aggregation [2, 14].

Let $\tilde{\mathbf{q}}_s$ and $\tilde{\mathbf{p}}_k$ denote the L_2 -normalized query and prototype, respectively. We compute the temperature-scaled similarity as $s_k = \langle \tilde{\mathbf{q}}_s, \tilde{\mathbf{p}}_k \rangle / \tau_{\text{ret}}$, where τ_{ret} is the retrieval temperature. We then select the top- k_{ret} prototypes according to s_k , with their index set denoted as $\mathcal{T}$. The context-conditioned guidance is then formed by

$$\mathbf{G}_{\text{guide}} = \sum_{k \in \mathcal{T}} \frac{\exp(s_k)}{\sum_{j=1}^K \exp(s_j)} \mathbf{p}_k. \quad (15)$$

Different from top- k re-normalization, the selected weights are normalized over the full prototype bank, and thus their total mass within $\mathcal{T}$ is no larger than one. Consequently, concentrated query-prototype matches provide stronger structural guidance, whereas uncertain or dispersed matches are naturally attenuated. For brevity, we denote the full-bank softmax weight as $\alpha_k = \exp(s_k) / \sum_{j=1}^K \exp(s_j)$ and use $c = \max_k \alpha_k$ as the top-1 routing confidence in bounded decoder modulation.

4.3.3 Prototype-Guided Coordination.

As illustrated in Figure 2, Prototype-Guided Coordination transforms the decoder state before reconstruction projection, $\mathbf{H}^{(0)}$, into $\mathbf{H}^{\text{dec}}$ using retrieved guidance. MaskAttention, comprising PAMG and PGMA, provides prototype-aware attention support [35], while Bounded FiLM performs confidence-aware, magnitude-bounded feature calibration [29].

Prototype-Aware Mask Generation (PAMG). Within MaskAttention, PAMG first converts the retrieved structural guidance into a prototype-aware support mask over decoder features. This

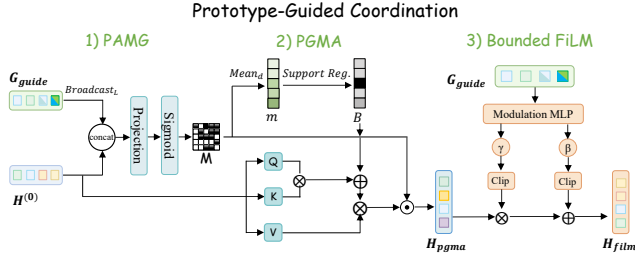


Figure 2: Prototype-guided coordination.

mask estimates where and to what extent the current decoder state is supported by the retrieved normal structural prototype. Let $\mathbf{H}^{(0)} \in \mathbb{R}^{L \times d}$ denote the decoder input, and let $\mathbf{G}_{\text{guide}} \in \mathbb{R}^d$ denote the retrieved guidance vector. As shown in Figure 2, $\mathbf{G}_{\text{guide}}$ is first broadcast along the temporal dimension to obtain $\tilde{\mathbf{G}}_{\text{guide}} \in \mathbb{R}^{L \times d}$, and then concatenated with $\mathbf{H}^{(0)}$. The concatenated representation is passed through a lightweight mask projection layer followed by a sigmoid activation:

$$\mathbf{M} = \sigma\left(\text{Proj}_m\left(\mathbf{H}^{(0)} \oplus \tilde{\mathbf{G}}_{\text{guide}}\right)\right), \quad (16)$$

where $\text{Proj}_m(\cdot)$ is a learnable linear projection parameterized by $\mathbf{W}_m$ and $\mathbf{b}_m$, i.e., $\text{Proj}_m(\mathbf{H}^{(0)} \oplus \tilde{\mathbf{G}}_{\text{guide}}) = [\mathbf{H}^{(0)} \oplus \tilde{\mathbf{G}}_{\text{guide}}] \mathbf{W}_m + \mathbf{b}_m$. The resulting $\mathbf{M} \in [0, 1]^{L \times d}$ provides a fine-grained guidance field that aligns the retrieved window-level structural prototype with token- and feature-level decoder states, serving as the basis for subsequent prototype-guided masked attention.

Prototype-Guided Masked Attention (PGMA). PGMA injects prototype guidance into decoder self-attention while keeping temporal dependency modeling in the main reconstruction branch. Given $\mathbf{M} \in [0, 1]^{L \times d}$, we obtain the token-level support score by feature-wise averaging, $m_j = \text{Mean}_d(\mathbf{M}_{j,:})$, and clip it as $\tilde{m}_j = \text{Clip}(m_j, \epsilon, 1 - \epsilon)$ for numerical stability. The ‘‘Support Reg.’’ operation in Fig. 2 maps the clipped support score into the attention-logit space as key-side structural guidance:

$$B_{ij} = \lambda_B [\log(\tilde{m}_j) - \log(1 - \tilde{m}_j)], \quad (17)$$

where $i, j = 1, \dots, L$ denote the query and key/value temporal positions, respectively, and λ_B controls the strength of prototype-support registration. The resulting key-side bias is then incorporated into decoder self-attention:

$$\text{Attn}(\mathbf{H}^{(0)}; \mathbf{B}) = \text{Softmax}_j \left(\frac{\mathbf{QK}^T}{\sqrt{d}} + \mathbf{B} \right) \mathbf{V}. \quad (18)$$

Adding B_{ij} reweights key/value tokens by their scaled prototype-support odds, while the query-key term preserves input-dependent temporal affinity.

The contextualized decoder response is then filtered by the original feature-level mask:

$$\mathbf{H}^{\text{pgma}} = \text{Attn}(\mathbf{H}^{(0)}; \mathbf{B}) \odot \mathbf{M}. \quad (19)$$

Thus, $\mathbf{B}$ guides where decoder self-attention aggregates temporal context, whereas $\mathbf{M}$ filters which feature dimensions are preserved.

Bounded FiLM. Following PGMA, Bounded FiLM calibrates the structurally guided decoder representation with constrained feature-wise modulation. Conditioned on the retrieved guidance vector

$\mathbf{G}_{\text{guide}}$, a modulation head produces raw scaling and shifting parameters $\gamma^{(0)}$ and $\beta^{(0)}$. To limit unreliable or excessive modulation, we convert them into confidence-aware bounded modulation factors, where $\gamma = 1 + \text{Clip}(c \tanh(\gamma^{(0)}), -\gamma_{\max}, \gamma_{\max})$ and $\beta = \text{Clip}(c \beta^{(0)}, -\beta_{\max}, \beta_{\max})$. Here, $c = \max_k \alpha_k$ is the top-1 routing confidence defined in Sec. 4.3.2, and $\gamma_{\max}$ and $\beta_{\max}$ bound the modulation magnitudes. These bounded factors are applied to the masked decoder feature $\mathbf{H}^{\text{pgma}}$:

$$\tilde{\mathbf{H}}^{\text{film}} = \gamma \odot \mathbf{H}^{\text{pgma}} + \beta. \quad (20)$$

We further compute the relative energy change induced by FiLM:

$$r = \frac{\text{Mean}\left(\left(\tilde{\mathbf{H}}^{\text{film}} - \mathbf{H}^{\text{pgma}}\right)^2\right)}{\text{Mean}\left(\left(\mathbf{H}^{\text{pgma}}\right)^2\right) + \epsilon}. \quad (21)$$

To avoid excessive perturbation of decoder features, the modulation is accepted only when $r \leq \rho$:

$$\mathbf{H}^{\text{dec}} = \begin{cases} \tilde{\mathbf{H}}^{\text{film}}, & r \leq \rho, \\ \mathbf{H}^{\text{pgma}}, & \text{otherwise.} \end{cases} \quad (22)$$

Since the retrieval confidence c scales the modulation factors, uncertain prototype matches naturally induce weaker decoder calibration. The energy constraint further rejects excessive feature perturbation even after modulation is generated.

4.4 Overall Objective

Following Eq. (1), ViCO optimizes the temporal detector with reconstruction learning and visual-to-temporal knowledge transfer. For a mini-batch $\mathcal{B} \subset \mathcal{W}_{\text{tr}}$, the decoder output is projected back to the input dimension and de-normalized before reconstructing each normal window as $\hat{\mathbf{X}}_i = F_\theta(\mathbf{X}_i; \mathcal{P})$. Meanwhile, the frozen visual teacher provides a teacher embedding $\mathbf{z}_i^{\text{tea}}$ for $\mathbf{X}_i$. Let $\mathbf{q}_{s,i}$ denote the student query embedding and $P_g(\cdot)$ denote a projection head that maps the student query into the teacher embedding space. The overall training objective is defined as

$$\mathcal{L}_{\text{train}} = \mathcal{L}_{\text{rec}} + \lambda_{\text{global}} \mathcal{L}_{\text{global}} + \lambda_{\text{rkd}} \mathcal{L}_{\text{rkd}}, \quad (23)$$

where λ_{global} and λ_{rkd} balance instance-level and relational visual-to-temporal transfer losses that complement reconstruction learning with structural supervision. $\mathcal{L}_{\text{global}}$ transfers instance-level visual semantics by aligning each student query with its teacher embedding. $\mathcal{L}_{\text{rkd}}$ transfers batch-level structural knowledge by preserving pairwise relations among normal samples [28]. The three loss terms are instantiated as

$$\begin{aligned} \mathcal{L}_{\text{rec}} &= \frac{1}{|\mathcal{B}|LC} \sum_{i \in \mathcal{B}} \|\mathbf{X}_i - \hat{\mathbf{X}}_i\|_F^2, \\ \mathcal{L}_{\text{global}} &= \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \left\| P_g(\mathbf{q}_{s,i}) - \mathbf{z}_i^{\text{tea}} \right\|_2^2, \\ \mathcal{L}_{\text{rkd}} &= \text{RKD} \left(\{P_g(\mathbf{q}_{s,j})\}_{j \in \mathcal{B}}, \{\mathbf{z}_j^{\text{tea}}\}_{j \in \mathcal{B}} \right). \end{aligned} \quad (24)$$

Here, $\mathcal{L}_{\text{rec}}$ learns normal temporal reconstruction, $\mathcal{L}_{\text{global}}$ aligns each student query with its visual teacher embedding, and $\mathcal{L}_{\text{rkd}}$ preserves the relational structure of teacher representations within the mini-batch. Specifically, $\text{RKD}(\cdot, \cdot)$ matches the pairwise cosine similarity matrices computed from projected student queries and teacher embeddings.

Algorithm 1: Prototype-Guided Training of ViCO

Input: normal training windows $\mathcal{W}_{tr}$; frozen visual teacher f_T ;
temporal detector F_θ ; prototype number K ; epoch number
 E ; learning rate η ; loss weights λ_{global} and λ_{rkd} .
Output: trained temporal detector $F_{\theta^*}(\cdot; \mathcal{P})$; prototype bank $\mathcal{P}$.
Notation: $\Phi_T(\cdot) = \text{MeanPool}(f_T(\text{TriView}(\cdot)))$.

- 1 **Normal Prototype Construction:**
- 2 $\mathcal{Z}_v \leftarrow \{\Phi_T(\mathbf{X}_i) \mid \mathbf{X}_i \in \mathcal{W}_{tr}\}$
- 3 $\mathcal{P} \leftarrow \text{KMeans}(\mathcal{Z}_v, K)$
- 4 **Prototype-Guided Student Optimization:**
- 5 Initialize trainable parameters θ
- 6 **for** each mini-batch $\mathcal{B} \subset \mathcal{W}_{tr}$ over E epochs **do**
- 7 $\mathbf{Z}_B^{tea} \leftarrow \Phi_T(\mathcal{B})$
- 8 $\theta \leftarrow \text{Student Update}(\mathcal{B}, \mathbf{Z}_B^{tea}, \theta, \mathcal{P})$
- 9 **Student Update** ($\mathcal{B}, \mathbf{Z}_B^{tea}, \theta, \mathcal{P}$):
- 10 $\mathbf{Q}_{s,B} \leftarrow \text{Query}_\theta(\mathcal{B})$
- 11 $\hat{\mathbf{X}}_B \leftarrow F_\theta(\mathcal{B}; \mathcal{P})$
- 12 $\mathcal{L} \leftarrow \mathcal{L}_{rec}(\mathcal{B}, \hat{\mathbf{X}}_B) + \lambda_{global} \mathcal{L}_{global}(\mathbf{Q}_{s,B}, \mathbf{Z}_B^{tea}) +$
- 13 $\lambda_{rkd} \mathcal{L}_{rkd}(\mathbf{Q}_{s,B}, \mathbf{Z}_B^{tea})$
- 14 $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$
- 15 **return** θ
- 16 **return** $F_{\theta^*}(\cdot; \mathcal{P})$ and $\mathcal{P}$

Algorithm 1 summarizes the optimization of Eq. (23): ViCO first constructs a fixed prototype bank from the frozen teacher representations and then updates the temporal detector using reconstruction and visual-to-temporal transfer losses. Here, $F_\theta(\mathcal{B}; \mathcal{P})$ denotes the prototype-guided forward pass introduced above.

At inference time, the visual teacher is no longer used. The trained temporal detector performs prototype-guided reconstruction using the fixed bank $\mathcal{P}$, and the anomaly score is computed from the reconstruction error according to Eq. (2).

5 Experiment

5.1 Experimental Setup

Datasets. We conduct experiments using 5 real-world datasets (MSL [13], SMAP [13], SMD [33], PSM [1], and GECCO [15]) to assess the performance of ViCO, covering server monitoring, satellite telemetry, and industrial systems.

Baselines. We compare ViCO with 16 baselines, including (1) Vision-augmented methods: TimeVLM [51], OccamVTS [23], and DMMV [32]; (2) Foundation time-series models: GPT4TS [52] and MOMENT [8]; (3) Dedicated TSAD methods: CrossAD [16], AnomalyTrans [40], CATCH [39], DCdetector [46], and TranAD [34]; (4) Generic temporal backbones: TimesNet [38], DLinear [47], ModernTCN [22], and FITS [44]; and (5) Classical methods: OCSVM [31] and LOF [6].

Metrics. Recent works have shown that point-adjustment protocols may cause unfairness [21, 30]. Therefore, we adopt the Volume Under the Surface of the ROC Curve (VUS-ROC) and the Range Area Under the Precision-Recall Curve (R-AUC-PR) as the

primary metrics in Table 1, since they provide robust, threshold-independent, range-aware evaluation. For comprehensive comparison, we additionally report Point-F1, Affiliation-F1 (Aff-F) [12], the Area Under the ROC Curve (AUC-ROC), the Area Under the Precision-Recall Curve (AUC-PR), the Range Area Under the ROC Curve (R-AUC-ROC) [27], and the Volume Under the Surface of the Precision-Recall Curve (VUS-PR) [3]. Together, these metrics assess point-wise, event-level and range-aware detection quality. Higher values indicate better performance.

Implementation Details. All methods are implemented under a unified benchmarking pipeline [38]. Baseline architectures and training configurations follow their published implementations, and We follow the default hyperparameter setting in the original paper or the associated code of baselines, enabling fair comparison. Unless otherwise specified, we use window length $L = 96$, stride $S = 96$, and batch size 32. For visual knowledge construction, we use MAE-Large as the default visual teacher. ViCO is trained with Adam optimizer at a learning rate of $1e-4$ on NVIDIA RTX 3090 GPUs.

5.2 Overall Performance

5.2.1 Main Results. We first evaluate ViCO on five real-world datasets against 16 competitive baselines, as reported in Table 1. Overall, ViCO achieves consistently strong performance across all five datasets, attaining the best results in most settings and remaining highly competitive in the others, with maximum relative gains of 7.0% in VUS-ROC and 9.0% in R-AUC-PR over the strongest baseline. Competing temporal and vision-augmented methods are effective on individual datasets but less consistent across domains, supporting the value of task-aligned visual guidance for TSAD.

As shown in Table 1, traditional methods, i.e., OCSVM and LOF, generally perform worse than deep learning methods, because complex temporal correlations are difficult to capture with their limited feature extraction capabilities. General temporal backbones, dedicated TSAD methods, foundation models, and vision-augmented baselines achieve competitive results on some datasets, but their gains are inconsistent across domains. This suggests that generic temporal modeling, pre-trained knowledge, or forecasting-oriented visual representations alone may be insufficient for robust TSAD, especially when anomaly patterns vary substantially across telemetry and industrial scenarios.

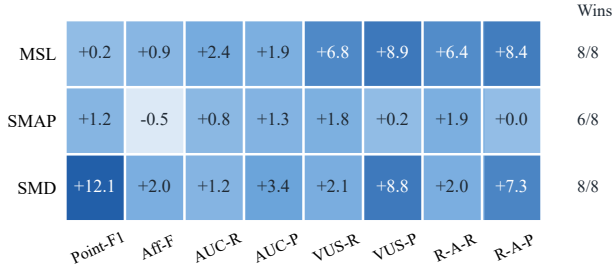
ViCO is particularly strong on MSL, whose heterogeneous anomalies benefit from complementary structural references. The results on PSM further highlight the importance of aligning external knowledge with the TSAD objective. On PSM, ViCO achieves the highest R-AUC-PR and remains close to OccamVTS in VUS-ROC. OccamVTS transfers low-level visual patterns from large vision models to lightweight temporal models for forecasting, which may help capture texture-like temporal structures. In contrast, ViCO retrieves and adapts normal structural prototypes to guide reconstruction-based anomaly scoring, where this task-aligned coordination yields stronger PR-oriented performance. On sparse-anomaly GECCO, this advantage is less pronounced and GPT4TS better exploits raw temporal dynamics.

5.2.2 Multi-metric Evaluation. We further evaluate ViCO under multiple protocols to examine whether its improvement is robust

Table 1: Results on the five real-world datasets. The best ones are in bold, and the second ones are underlined.

Method	MSL		SMD		SMAP		PSM		GECCO	
	VUS-ROC	R-AUC-PR	VUS-ROC	R-AUC-PR	VUS-ROC	R-AUC-PR	VUS-ROC	R-AUC-PR	VUS-ROC	R-AUC-PR
OCSVM	0.3980	0.1040	0.5390	0.0791	0.5157	0.1434	0.4515	0.3417	0.7079	0.0440
LOF	0.5186	0.1324	0.4409	0.0538	0.5324	0.1472	0.3964	0.3043	0.5822	0.0278
DLinear	0.6165	0.1834	0.7607	0.1750	0.4021	0.1427	0.5764	0.4482	0.9397	0.4200
FITS	0.6172	0.1868	0.7670	0.1844	0.5058	0.1382	0.6222	0.4654	0.9750	0.5071
TimesNet	<u>0.6627</u>	<u>0.2153</u>	0.8164	0.2108	0.5193	0.1583	0.5940	0.4497	0.9918	0.6542
ModernTCN	0.6157	0.1818	0.7573	0.1855	0.5260	0.1388	0.6136	0.4602	0.9797	0.5376
TranAD	0.6161	0.1877	0.7073	0.1528	0.4305	0.1192	0.5005	0.3999	0.9521	0.4943
AnomalyTrans	0.6590	0.2144	0.8132	0.1990	0.5106	0.1586	0.6000	0.4461	0.9920	0.6570
DCdetector	0.6504	0.2131	0.8132	0.2012	0.5153	0.1538	0.5877	0.4320	0.9902	0.5939
CrossAD	0.6339	0.1979	0.7006	0.1459	0.4698	0.1318	0.6075	0.4779	0.9904	0.6246
CATCH	0.6576	0.2137	0.8130	0.2146	0.5279	<u>0.1643</u>	0.6231	0.4894	0.9791	0.4946
GPT4TS	0.6405	0.1869	0.8016	0.1991	0.5247	0.1395	0.6231	0.4861	0.9932	0.7704
MOMENT	0.6366	0.1966	0.8113	<u>0.2155</u>	0.5240	0.1381	0.6271	0.4852	0.9894	0.6378
TimeVLM	0.6295	0.2035	<u>0.8191</u>	0.2127	0.5463	0.1471	0.6339	0.4825	0.9896	0.6331
OccamVTS	0.6442	0.2099	0.8112	0.2066	0.5483	0.1525	0.6394	<u>0.4954</u>	0.9882	0.6141
DMMV	0.6563	0.2103	0.8118	0.2095	<u>0.5674</u>	0.1477	0.6242	0.4716	0.9919	0.6559
ViCO	0.7090	0.2347	0.8365	0.2318	0.5791	0.1717	<u>0.6348</u>	0.4983	<u>0.9926</u>	<u>0.6768</u>

to different evaluation criteria. This analysis is conducted on MSL, SMAP, and SMD, which cover different anomaly characteristics, including point-wise, contextual, and sequence-level collective anomalies. We consider eight metrics spanning point-wise accuracy, event-level affiliation, threshold-independent ranking, and range-aware evaluation, thereby reducing reliance on any single scoring protocol. For each dataset-metric pair, Figure 3 reports the relative gain of ViCO over the strongest competing baseline under the corresponding metric.

**Figure 3: Results of multi-metric evaluation.**

As shown in Figure 3, ViCO ranks first or ties for first in 23 of 24 comparisons, with an average improvement of 3.4%. The gains are particularly consistent on MSL and SMD, where ViCO ranks first or ties for first under all eight metrics, indicating that the proposed coordination mechanism improves detection quality from multiple evaluation perspectives. On SMAP, ViCO also achieves positive gains on most metrics, with the only underperformance occurring on Aff-F, where it is slightly less favorable under the event-level affiliation criterion. Overall, these results indicate that ViCO’s improvement is not driven by a specific anomaly type or evaluation metric; rather, it remains stable across point-wise, event-level, threshold-independent, and range-aware protocols, further supporting the effectiveness of the proposed two-phase visual-to-temporal knowledge transfer design.

5.3 Anomaly-Sensitive Error Analysis

Rather than claiming a formal guarantee, we empirically examine whether ViCO’s vision augmentation improves anomaly scoring rather than merely reconstruction fidelity. By transferring normal structural knowledge from the tri-view visual branch to temporal reconstruction, ViCO is expected to enlarge the anomaly-normal residual gap while suppressing high residuals on normal samples. We therefore compare the reconstruction-error profile of full ViCO with that of the temporal student.

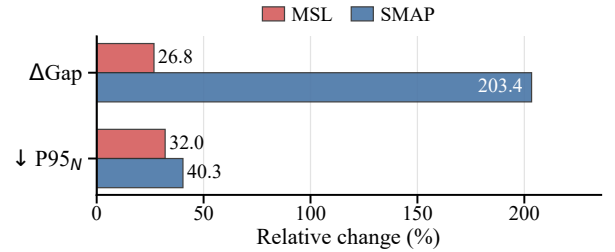
**Figure 4: Anomaly-sensitive error analysis.**

Figure 4 reports two diagnostics relative to the temporal student on MSL and SMAP. ΔGap denotes the relative increase in the anomaly-normal residual gap, and $\downarrow \text{P95}_N$ denotes the relative reduction of the 95th-percentile reconstruction error on normal samples. Positive values are better for both diagnostics.

ViCO improves both diagnostics: it increases ΔGap by 26.8% and 203.4% on MSL and SMAP, respectively, while reducing P95_N by 32.0% and 40.3%. These results show that vision augmentation makes abnormal deviations more distinguishable in the reconstruction-error space while suppressing high-score normal residuals, providing empirical evidence for more reliable anomaly scoring.

5.4 Ablation Study

We conduct ablation studies to assess the contributions of visual knowledge transfer and prototype-guided coordination to temporal

reconstruction. Specifically, we compare the full ViCO with five variants that remove visual knowledge construction, prototype-bank retrieval, relational distillation, or decoder-side coordination modules. *w/o Teach.*: ViCO without visual-teacher knowledge, where the temporal student is trained only with the reconstruction objective; *w/o Proto.*: ViCO without prototype-bank guidance, where visual knowledge is transferred only through distillation losses without retrieving normal structural prototypes for reconstruction; *w/o RKD*: ViCO without relational knowledge distillation, where the model removes batch-level structural relation preservation while retaining prototype-guided reconstruction; *w/o MAtt*: ViCO without the MaskAttention mechanism, i.e., removing PAMG and PGMA and injecting retrieved guidance only through Bounded FiLM; *w/o FiLM*: ViCO without Bounded FiLM, using only PAMG and PGMA for prototype-guided structural filtering.

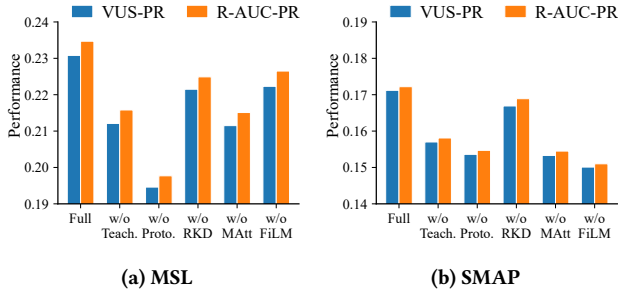


Figure 5: Ablation results.

Figure 5 reports VUS-PR and R-AUC-PR on MSL and SMAP. ViCO consistently outperforms all partial variants. Among the guidance-path variants, *w/o RKD* consistently performs better than *w/o Proto.*, indicating that prototype retrieval provides more effective reconstruction guidance than teacher supervision alone. Among the decoder-modulation variants, *w/o FiLM* outperforms the *w/o MAtt* variant on MSL by 5.1% in VUS-PR and 5.3% in R-AUC-PR, suggesting that PAMG and PGMA provide stronger local structural guidance than feature-wise modulation alone. Nevertheless, the full model still achieves the best performance, indicating that mask-biased local filtering and bounded feature-wise modulation are complementary.

5.5 Tri-view Effectiveness Analysis

We further analyze the visual construction side of ViCO to verify whether Tri-view rendering provides more effective normal structural references than alternative time-series-to-image designs. Specifically, we first compare different rendering strategies under the same reconstruction setting, and then inspect representative cases to understand how different views expose complementary anomaly cues.

5.5.1 Rendering Strategy Comparison. Table 2 reports the results of LinePlot rendering, single-view variants, two-view combinations, and the complete Tri-view design under the same reconstruction-phase setting. Tri-view achieves the best performance on both datasets, improving the two PR-oriented range-aware metrics over LinePlot by 7.3%–7.6% on MSL and 10.6%–10.8% on SMAP. This indicates that directly rendering raw trajectories is not sufficient for

constructing effective structural references, while reorganizing time series from multiple structural perspectives provides more useful visual knowledge for reconstruction-based anomaly detection.

Table 2: Comparison of visual rendering strategies.

Variant	MSL		SMAP	
	VUS-PR	R-AUC-PR	VUS-PR	R-AUC-PR
LinePlot	0.2151	0.2181	0.1549	0.1553
STFT	0.2160	0.2191	0.1625	0.1637
Heatmap	0.2174	0.2205	0.1614	0.1622
Gradient	0.2204	0.2242	0.1618	0.1629
STFT+Heatmap	0.2228	0.2266	0.1677	0.1694
STFT+Gradient	0.2147	0.2182	0.1642	0.1650
Heatmap+Gradient	0.2177	0.2212	0.1687	0.1702
ViCO	0.2308	0.2347	0.1717	0.1717

The comparison among different view configurations further shows that each view contributes differently across datasets. Among single-view variants, Gradient performs best on MSL, suggesting that local-transition cues are informative for this telemetry dataset. On SMAP, Heatmap-related combinations are strong, indicating that contextual value organization is helpful for capturing its anomaly patterns. However, no single-view or two-view variant consistently matches the complete Tri-view design. These results support the complementarity of STFT, Heatmap, and Gradient, showing that the three views provide non-redundant structural cues for constructing anomaly-oriented visual knowledge.

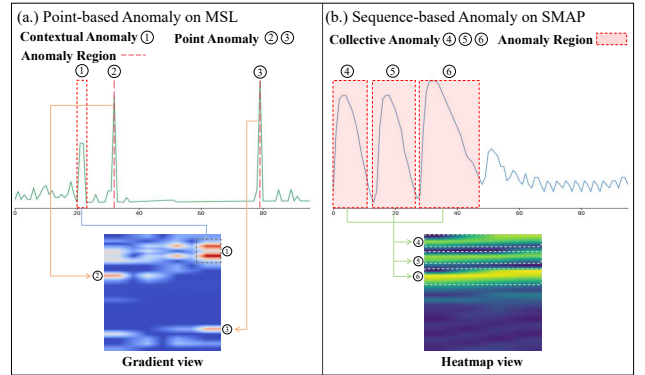


Figure 6: Anomaly-view alignment examples.

5.5.2 Case Study. To further inspect how Tri-view rendering exposes such complementary cues, we visualize representative cases from MSL and SMAP in Figure 6. On MSL, the Gradient view highlights point-based anomalies as narrow high-response bands and captures contextual anomalies as broader local response regions, which is consistent with the strong quantitative performance of the Gradient view in Table 2. On SMAP, the Heatmap view makes sequence-level anomalies more explicit by revealing sustained trajectory distortions over contiguous intervals. These cases provide intuitive evidence that different views emphasize different anomaly-related structures, explaining why their combination yields more robust visual references than any individual view.

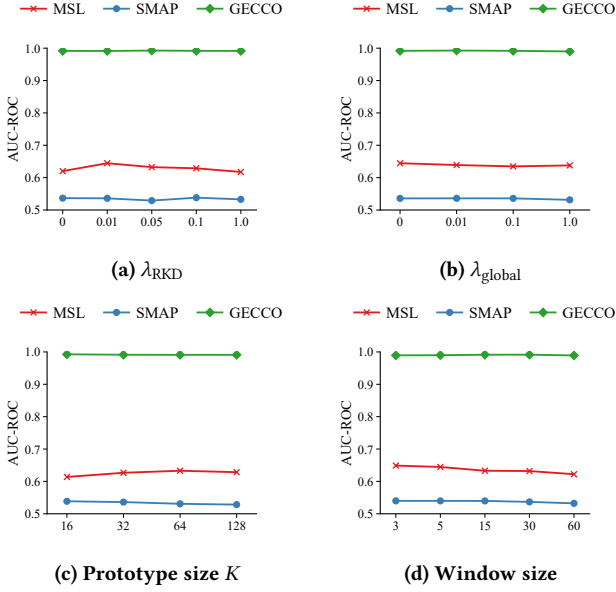


Figure 7: Parameter sensitivity.

5.6 Sensitivity and Efficiency Analysis

We further examine ViCO from the perspectives of hyperparameter robustness and inference efficiency.

5.6.1 Parameter Sensitivity. Figure 7 reports the AUC-ROC under different λ_{RKD} , λ_{global} , prototype size K , and smoothing window size on MSL, SMAP, and GECCO. First, ViCO is generally robust to these hyperparameters: small transfer weights are preferred, while overly large λ_{global} can weaken performance. Second, the prototype bank does not require a large size. Compact prototype banks are sufficient, and larger K yields limited or negative gains. Third, score smoothing uses a centered moving average with edge padding after timeline alignment. For Table 1, the fixed windows are 5, 15, 30, 37, and 21 for MSL, SMD, SMAP, PSM, and GECCO, respectively. The curves show that small windows preserve local variation, whereas large windows oversmooth anomaly scores. Overall, ViCO performs reliably with small transfer weights, compact prototype banks, and moderate smoothing.

5.6.2 Inference Efficiency. We evaluate inference efficiency of ViCO on MSL. Figure 8 reports peak GPU memory usage and per-batch

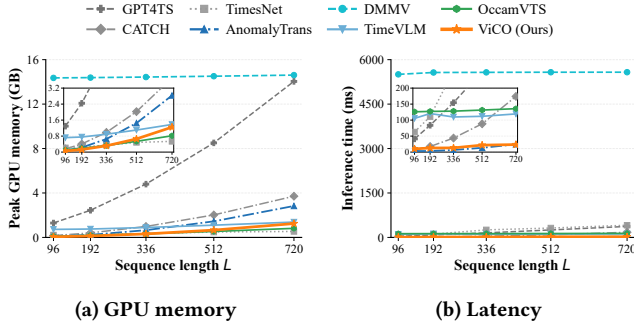


Figure 8: Efficiency analysis.

Table 3: Efficiency comparison on MSL at $L = 720$. * reports the last feasible result at $L = 512$ due to OOM.

Type	Method	VUS-ROC	Train. (M)	Total (M)	Mem. (GB)	Infer. (ms)
Generic temporal	TimesNet	0.6604	37.52	37.52	0.54	410.15
	DLinear	0.6165	0.02	0.02	0.05	0.61
	ModernTCN	0.6157	72.72	72.72	0.91	29.97
Foundation TS	GPT4TS	0.6405	1.18	124.79	14.04	371.01
	MOMENT	0.6370	0.01	109.64	7.76	2150.92
Dedicated TSAD	CrossAD	0.6339	0.75	0.75	5.84	59.76
	CATCH	0.6639	14.95	14.95	3.71	174.01
	AnomalyTrans	0.6590	0.29	0.29	2.83	24.36
	DCdetector	0.6504	0.15	0.15	14.86*	78.95*
	TranAD	0.6161	0.26	0.26	0.69	8.27
Vision-augmentation	TimeVLM	0.6295	0.75	0.75	1.38	119.31
	OccamVTS	0.6442	0.63	0.63	0.82	135.69
	DMMV	0.6563	0.71	0.71	14.61	5577.18
ViCO		0.7090	0.40	0.40	1.25	23.51

latency as input length increases, while Table 3 presents the full accuracy–efficiency trade-off. ViCO maintains stable memory use and low inference latency, achieving the best VUS-ROC of 0.7090 with only 0.40M trainable parameters, 1.25 GB memory, and 23.51 ms inference latency. Although some lightweight baselines require fewer resources, they show clearly weaker detection performance; in contrast, foundation-model and visual baselines often introduce larger memory or latency costs without achieving better accuracy. This balance stems from ViCO’s teacher-free inference design. Visual rendering and MAE encoding are confined to offline knowledge construction, whereas test-time detection uses the temporal student and fixed prototype-bank retrieval only. The prototype bank therefore acts as lightweight retrieval knowledge rather than a test-time visual branch.

6 Conclusion

This work presents ViCO, a vision-augmented coordination framework for time series anomaly detection. By modeling normal behavior from visual-structural and temporal-value perspectives and coordinating retrieved structural prototypes with temporal decoding, ViCO yields more discriminative reconstruction errors and achieves efficient teacher-free detection. An interesting research direction is to further improve the proposed ViCO to fit streaming time series in non-stationary environments.

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GenAI Usage Disclosure

The authors used generative AI tools only for language polishing and grammar checking. Authors take full responsibility for the content of this paper.

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